



Dr. Fatih Nayebi is a seasoned expert in AI and Human-Computer Interaction. He holds a PhD in Engineering from École de technologie supérieure, specializing in Machine Learning and HCI, and completed a post-doctoral fellowship in Machine Learning.

He has leveraged this deep technical background to drive innovation in the retail industry. Currently, Dr. Nayebi serves as the VP of Data & AI at the ALDO Group, a leading global retailer, where he spearheads data-driven strategies and the development of intelligent retail solutions.

In addition to his industry leadership, Dr. Nayebi has been a Faculty Lecturer at McGill University for the past six years, teaching courses such as Enterprise Data Science, Machine Learning in Production, and Deep Learning.

He was an early pioneer in applied AI—bringing his first productionized AI product to life in 2008—and continues to be at the forefront of the field. He is also the author of the Swift Functional Programming books, demonstrating his passion for software development and sharing knowledge. Dr. Nayebi's unique blend of academic insight and real-world experience positions him as a thought leader in the intersection of AI and retail.

What makes this book indispensable?

- **75+ diagrams, flowcharts, and architecture blueprints:** rendered in vivid color, making complex concepts, data flows, and decision loops crystal-clear.
- **50+ Real World Retail Use Cases:** Explore detailed blueprints and discussions covering demand forecasting, dynamic pricing, conversational merchandising, autonomous store operations, supply chain optimization, and many more—illustrating agent capabilities across the retail landscape.
- **28 Production Ready Code Examples:** Get hands-on with complete, copy-and-run Python listings covering: BDI agents, OODA loops, Reinforcement Learning pipelines, LLM-powered ReAct chains, MCP-based negotiations, A2A multi-agent orchestration, and beyond.
- **Retail Focused, Industry Agnostic Foundations:** Master the core math, decision frameworks (like MDPs, Bayesian methods), and reference architectures applicable to any industry, then dive deep into specialized adaptations proven in retail.
- **Traditional Rigor + Cutting Edge Tech:** Bridge foundational AI (optimization, planning) with the latest breakthroughs: Large Language Models (LLMs), transformer agents, retrieval-augmented generation (RAG), and modern open multi-agent protocols like Anthropic's MCP and Google's A2A.

Foundations of Agentic AI for Retail

This book is your ultimate playbook for thriving in a rapidly evolving retail landscape. Written by a leading retail AI pioneer, this authoritative guide uncovers a powerful new paradigm of intelligent systems that don't just process data, but actively think, adapt, and learn in real time.

Inside, you'll discover how Agentic AI is already reshaping everything from supply chain operations and inventory management to omnichannel personalization and customer engagement. Drawing upon in-depth frameworks, real code examples, and hands-on case studies, this book provides a clearroadmap for designing, deploying, and orchestrating autonomous systems that deliver tangible business results. Whether you're a retail executive aiming to future-proof your strategy, a tech professional eager to implement next-generation solutions, or an AI enthusiast curious about the next frontier of machine intelligence, Foundations of Agentic AI for Retail shows you how to:

- **Reimagine Retail Operations:** Gain unprecedented visibility and control over every step of the retail value chain, using Agentic AI to optimize processes, reduce costs, and improve speed-to-market.
- **Elevate the Customer Experience:** Leverage predictive analytics, real-time personalization, and cutting-edge conversational agents to delight shoppers at every touchpoint—online, in-store, and beyond.
- **Implement with Confidence:** Navigate the technical architecture, integration challenges, and ethical considerations that shape real-world AI deployments, guided by practical code snippets and proven best practices.
- **Master the Multi-Agent Future:** Discover how Agentic AI collaborates across functions—pricing, marketing, fulfillment, and more—to deliver a seamless, data-driven ecosystem that propels your retail business ahead of the competition.



Foundations of Agentic AI for Retail

Concepts, Technologies, and Architectures for Autonomous Retail Systems



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